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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

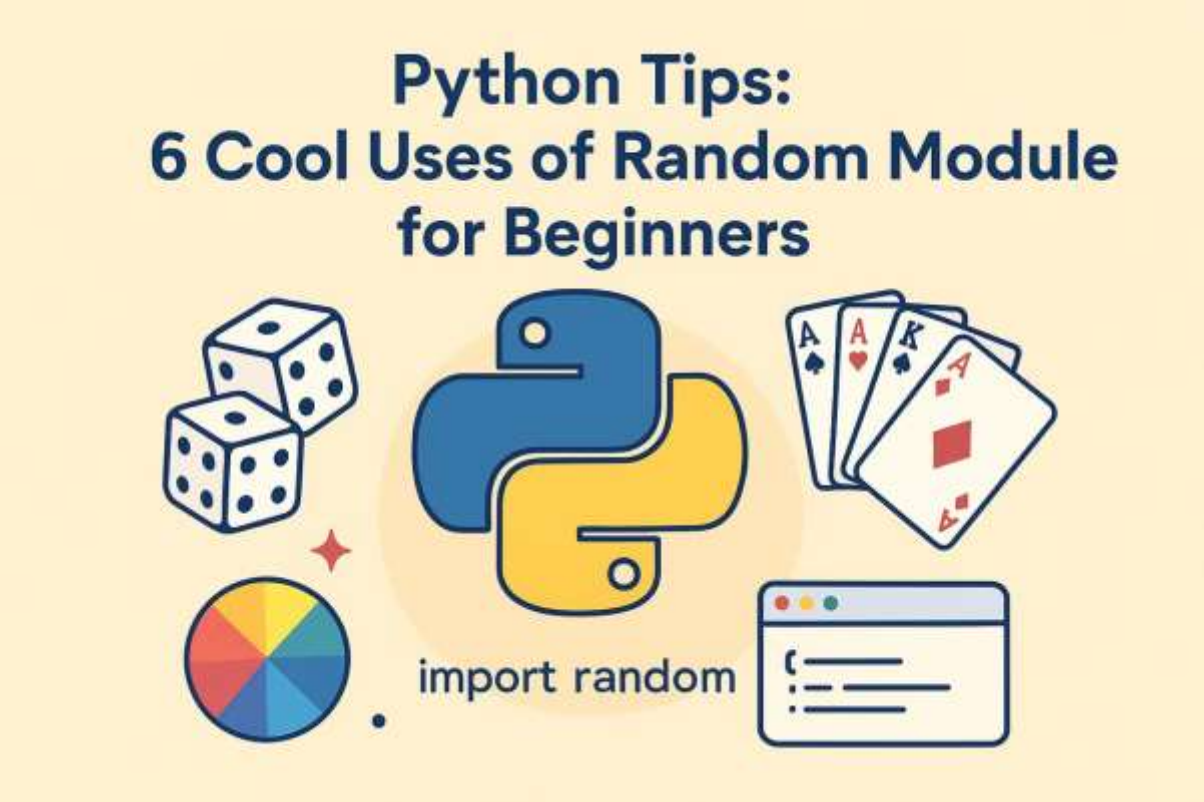
Classes : Offline & Online

Daily Teaching : 2 Hours

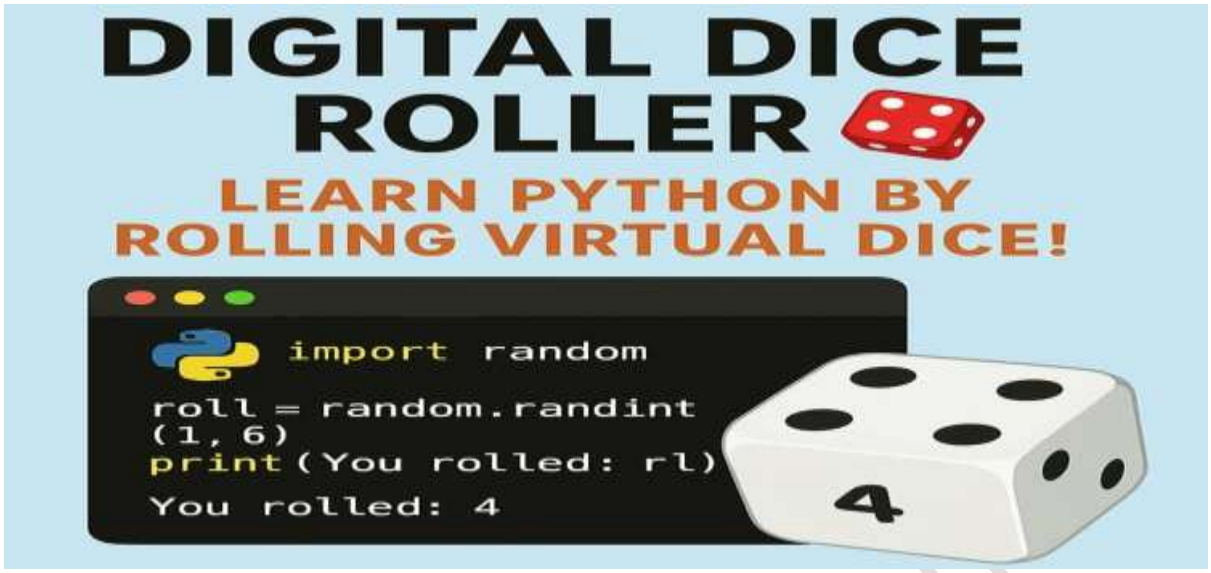
Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Class 34: Random Module in Python



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Learning Objectives

By the end of this lesson, students will be able to:

- Understand what a **module** is.
- Understand the **Random Module**.
- Generate random numbers.

- Select random items from a list.
- Shuffle items randomly.
- Create simple games using random numbers.
- Build a basic **Guess the Number Game**.

1. What is a Module?

A **module** is a Python file that contains useful code, functions, and tools that we can use in our programs.

Think of a module like a **toolbox** .

Instead of creating every tool ourselves, Python gives us many ready-made tools.

Example

Python has modules such as:

- random
- math
- datetime
- turtle

Today we are learning about the **random module**.

2. What is the Random Module?

The **random module** is a built-in Python module used to generate random values and make random choices.

For example, we can use it to:

- Generate a random number
- Pick a random fruit
- Roll a virtual dice
- Toss a virtual coin
- Shuffle a list
- Create simple games

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3. How to Import the Random Module

Before using the random module, we normally import it.

import random

Meaning:

import → Bring a module into our program.

random → Name of the module.

So:

import random

means:

"Python, please give me the tools from the random module."

4. Generate a Random Number

We can use randint() to generate a random whole number between two numbers.

Syntax

random.randint(start, end)

Example

import random

number = random.randint(1, 10)

print(number)

Possible output:

7

Another time it might be:

3

Another time:

10

The result can change each time the program runs.

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Important

random.randint(1, 10)

can generate:

1, 2, 3, 4, 5, 6, 7, 8, 9, or 10

Both the starting and ending numbers are included.

5. Virtual Dice

A dice normally has numbers from **1 to 6**.

We can create a virtual dice using randint().

```
import random

dice = random.randint(1, 6)

print("Dice:", dice)
```

Possible output:

Dice: 4

Run it again:

Dice: 2

Run it again:

Dice: 6

🎲 This is how computers can simulate a dice roll.

6. Coin Toss

We can create a simple coin toss.

```
import random

coin = random.choice(["Heads", "Tails"])
```

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print(coin)

Possible output:

Heads

or:

Tails

7. random.choice()

choice() selects **one random item** from a list.

Syntax

random.choice(list)

Example

import random

fruits = ["Apple", "Banana", "Mango", "Orange"]

fruit = random.choice(fruits)

print(fruit)

Possible output:

Mango

The next time it may select:

Apple

8. Random Choice from Games

We can use choice() to randomly select a game.

import random

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```
games = ["Cricket", "Football", "Chess", "Badminton"]

game = random.choice(games)

print("Today's game:", game)
```

Possible output:

Today's game: Chess

9. random.shuffle()

shuffle() changes the order of items in a list randomly.

Example

```
import random

numbers = [1, 2, 3, 4, 5]

random.shuffle(numbers)

print(numbers)
```

Possible output:

[3, 1, 5, 2, 4]

Another run may produce:

[5, 2, 1, 4, 3]

Before and After Shuffle

Before:

1 2 3 4 5

After:

4 1 5 2 3

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The items are still there, but their order changes.



```python
import random

cards = ['Ace' , 'King' , 'Queen' , 'Jack']
# Shuffle the list
random.shuffle(cards)
print(cards)
```

```python
# Import the random module
import random

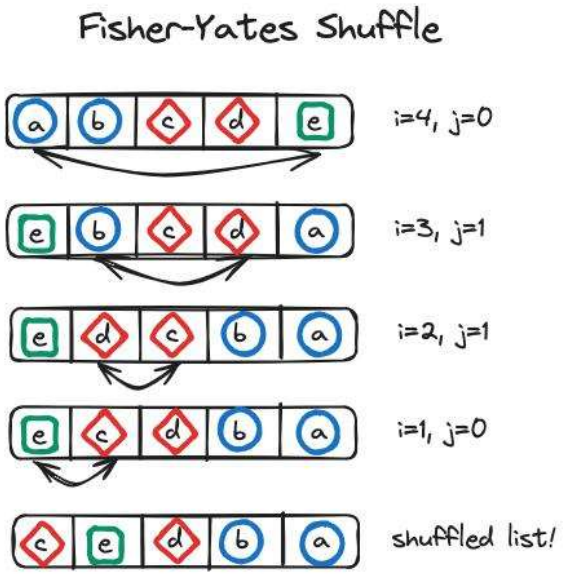
# Define a list
sample_list = ['A', 'B', 'C', 'D', 'E']

# Shuffle the list in-place
random.shuffle(sample_list)

# Print the shuffled list
print("After the first shuffle:", sample_list)

# Shuffle the list again
random.shuffle(sample_list)

# Print the shuffled list again
print("After the second shuffle:", sample_list)
```



10. random.random()

random.random() generates a random decimal number from **0.0 up to but not including 1.0**.

Example:

```
import random


number = random.random()


print(number)
```

Possible output:

0.735421

Another output could be:

0.182934

For 7th–8th standard students, remember:

random.random() gives a random decimal value between 0 and 1.

11. random.randrange()

randrange() selects a random number from a range.

```
import random
```

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number = random.randrange(1, 10)

print(number)

Possible output:

6

Important Difference

random.randint(1, 10)

can include **10**.

But:

random.randrange(1, 10)

does **not** include 10.

It chooses from:

1 2 3 4 5 6 7 8 9

Important Random Functions

Function	Purpose	Example
randint()	Random integer including both ends	randint(1,10)
choice()	Select one random item	choice(fruits)
shuffle()	Randomly rearrange a list	shuffle(numbers)
random()	Random decimal from 0 to <1	random()
randrange()	Random number from a range	randrange(1,10)

Practical Session

Practical 1: Random Number

import random

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```
number = random.randint(1, 100)

print("Random Number:", number)
```

This generates a random number from **1 to 100**.

Practical 2: Dice Game

```
import random

dice = random.randint(1, 6)

print("You rolled:", dice)
```

Sample Output

You rolled: 5

Practical 3: Coin Toss

```
import random

result = random.choice(["Heads", "Tails"])

print("Coin:", result)
```

Practical 4: Random Fruit

```
import random

fruits = ["Apple", "Banana", "Mango", "Orange"]

fruit = random.choice(fruits)
```

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```
print("Random Fruit:", fruit)
```

Practical 5: Random Student

```
import random

students = ["Ali", "Asha", "Rahul", "Sara"]

student = random.choice(students)

print("Selected Student:", student)
```

This can be useful for selecting a student randomly for a classroom activity.

Practical 6: Shuffle Students

```
import random

students = ["Ali", "Asha", "Rahul", "Sara"]

random.shuffle(students)

print("New Order:")
print(students)
```

Mini Project

Mini Project 1: Guess the Number Game

This is a great beginner project.

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The computer chooses a random number, and the player tries to guess it.

```
import random

secret = random.randint(1, 10)

guess = int(input("Guess a number from 1 to 10: "))

if guess == secret:
    print("🎉 Correct! You won!")
else:
    print("❌ Wrong!")
    print("The number was:", secret)
```

Example

```
Guess a number from 1 to 10: 7

🎉 Correct! You won!
```

Mini Project 2: Improved Guessing Game

```
import random

secret = random.randint(1, 10)

guess = int(input("Guess a number from 1 to 10: "))

if guess == secret:
    print("🎉 Excellent! You guessed correctly!")
elif guess < secret:
    print("Too low!")
```

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```
    print("The answer was:", secret)
else:
    print("Too high!")
    print("The answer was:", secret)
```

Mini Project 3: Dice Rolling Game

```
import random

print("🎲 Welcome to Dice Game!")

dice = random.randint(1, 6)

print("You rolled:", dice)

if dice == 6:
    print("🎉 Great! You got 6!")
else:
    print("Try again!")
```

Mini Project 4: Coin Toss Game

```
import random

print("🪙 Coin Toss Game")

choice = input("Choose Heads or Tails: ")

result = random.choice(["Heads", "Tails"])
```

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```
print("Computer:", result)


if choice.lower() == result.lower():
    print("🎉 You won!")
else:
    print("❌ You lost!")
```

Mini Project 5: Random Food Picker

```
import random


foods = [
    "Pizza",
    "Burger",
    "Biryani",
    "Dosa",
    "Noodles"
]


food = random.choice(foods)


print("Today's food suggestion:", food)
```

Classroom Practical Activity

Activity: "Computer Chooses a Student"

Step 1

Create a list:

```
students = ["Asha", "Rahul", "Ali", "Sara", "John"]
```

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Step 2

Import random:

import random

Step 3

Choose a student:

selected = random.choice(students)

Step 4

Display:

print("Today's selected student:", selected)

🎯 Ask students to run the program several times and observe how the selected student changes.

Challenge Activity

Create a list containing:

Rock

Paper

Scissors

Use random.choice() to make the computer select one.

import random

choices = ["Rock", "Paper", "Scissors"]

computer = random.choice(choices)

print("Computer chose:", computer)

This is the beginning of a **Rock-Paper-Scissors game**! 🎮

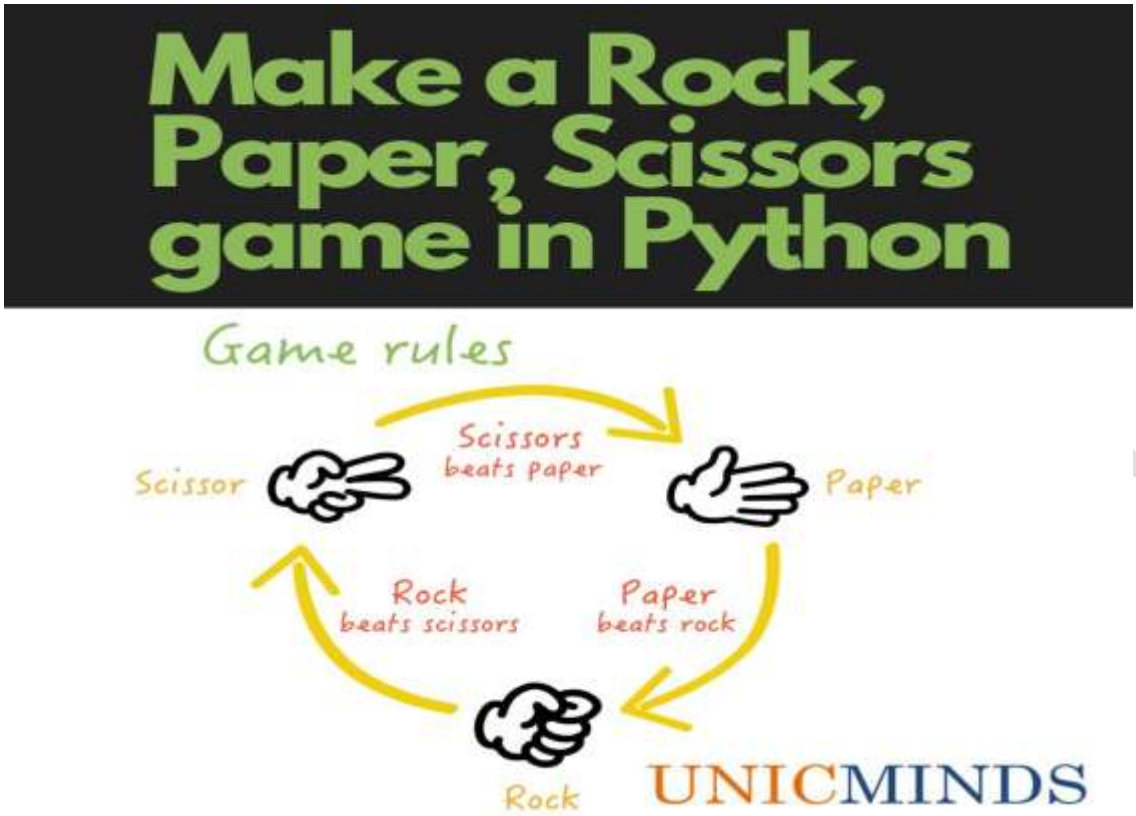
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PYTHON GAME: ROCK, PAPER, SCISSORS



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Common Mistakes

❌ **Mistake 1: Forgetting import**

Wrong:

```
number = random.randint(1, 10)
```

Correct:

```
import random

number = random.randint(1, 10)
```

❌ **Mistake 2: Incorrect spelling**

Wrong:

```
random.rantint(1, 10)
```

Correct:

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random.randint(1, 10)

❌ **Mistake 3: Forgetting brackets**

Wrong:

random.choice

Correct:

random.choice(["Red", "Blue", "Green"])

Important Difference

randint()

random.randint(1, 5)

Possible numbers:

1, 2, 3, 4, 5

randrange()

random.randrange(1, 5)

Possible numbers:

1, 2, 3, 4

⭐ Remember: randrange() excludes the ending number.

Student Practice Questions

Very Short Questions

1. What is a module?
2. What is the random module?
3. How do we import the random module?
4. What does randint() do?
5. What does choice() do?
6. What does shuffle() do?

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7. What does random() return?
8. What is the difference between randint() and randrange()?

Coding Questions

Question 1

Generate a random number between 1 and 50.

Question 2

Create a virtual dice.

Question 3

Create a coin toss program.

Question 4

Select a random color from:

["Red", "Blue", "Green", "Yellow"]

Question 5

Select a random student from a list of five students.

Question 6

Shuffle this list:

numbers = [1, 2, 3, 4, 5]

Homework

Create a "Lucky Number Generator"

Your program should:

1. Import the random module.
2. Ask the user for their name.
3. Generate a random number between **1 and 100**.
4. Display the student's lucky number.

Hint:

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```
import random

name = input("Enter your name: ")

number = random.randint(1, 100)

print(name, "your lucky number is:", number)
```

Lesson Summary

Today we learned:

- 📦 **Module** → A collection of useful Python code.
- 🎲 **Random Module** → Used for random values and choices.
- import random → Imports the random module.
- randint() → Generates a random integer.
- choice() → Selects one random item.
- shuffle() → Randomly rearranges a list.
- random() → Generates a decimal from 0 to less than 1.
- randrange() → Generates a random number from a range.

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